School of Future Tech

Case Study Report

Online Movie Ticket Booking

by

Atharva Ashok Dalvi

150096725179

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Introduction

This project simulates an online movie ticket booking system using C++.
It allows users to select movies, choose theatres, pick show timings, view seat availability, and book tickets. The system also generates a ticket and stores booking details in a file for record-keeping.

Problem Statement

Design and implement a console-based movie ticket booking system that enables users to:

- Select a movie from available options
- Choose a theatre where the movie is playing
- Select a preferred showtime
- View available seats
- Book seats and generate a ticket
- Store booking details permanently

Methods & Algorithms / Technology Used

- **Programming Language:** C++
- **Concepts Used:**
    - Multi-dimensional arrays for seat management
    - Conditional statements for validation

    - Loops for menu-driven system
    - File handling using `ofstream` to store bookings
- **Algorithm:**
    - Display movie list
    - Filter theatres based on movie availability
    - Show time slots
    - Display seat matrix
    - Accept user input (name, seat)
    - Validate input
    - Book seat if available
    - Print ticket and save data

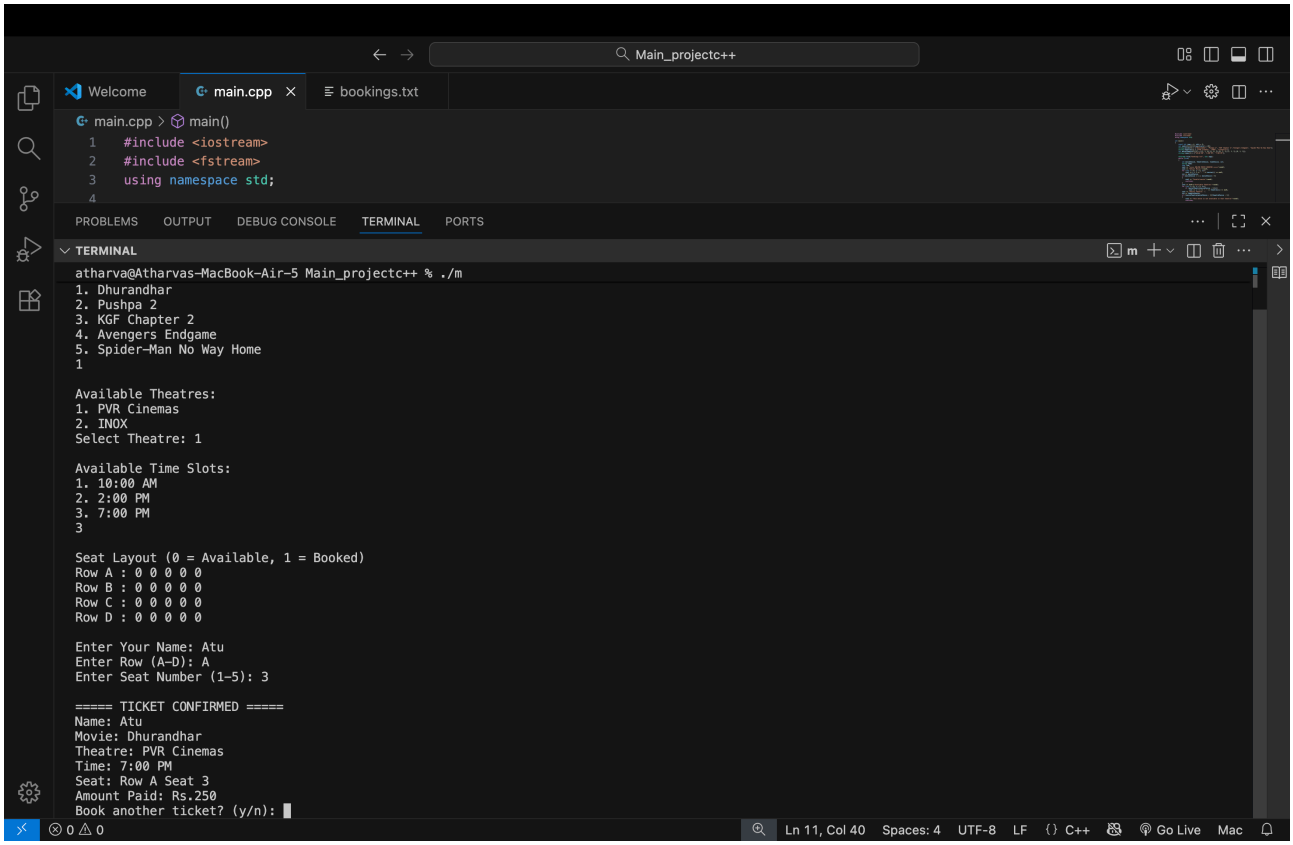
Implementation Details

- A **5D array** is used:

  `seats[movie][theatre][time][row][column]`

  to track seat availability.
- Movies, theatres, and timings are stored in arrays.
- `movieTheatre` matrix ensures only valid theatre options are shown.
- Seat layout uses:
    - `0` → Available
    - `1` → Booked
- File handling:
    - Booking details are saved in **bookings.txt** using append mode.

OUTPUT



Results and Conclusion

The system successfully simulates a real-world movie ticket booking process.
It ensures:

- Accurate seat tracking
- Prevention of duplicate bookings
- User-friendly interaction
- Persistent storage of booking data